

Title : Normalization

Category : Preprocessing

1- Objective :

Normalization makes EEG channels comparable across a session by putting the data on the same scale. In this method, the signal is standardized using z-score normalization, so each channel has zero mean and unit variance.

2- How it works :

The method works per session and per channel.

For each channel, it computes:

- mean, standard deviation

Then each EEG value is transformed using:

$$Z_i = (X_i - \text{mean}) / \text{standard deviation}$$

So, Every sample is recalculated using the channel mean and standard deviation from that session.

3- Example :

If one channel in a session has values: [2, 4, 6]

Mean: 4, Standard deviation: 1.63

Then the normalized values become: [-1.23, 0, 1.23]

So the signal keeps its shape, but its scale is standardized.